



MISSION 02 / CAD

Glass Bottle - Surface & Revolve Modelling

Recreate the reference bottle as a clean, manufacturable 3D CAD model.

REFERENCE IMAGE

Use the supplied bottle image as the visual reference for the overall form, proportions, neck transition, shoulder, body, base, and opening.



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YOUR TASK

Create a 3D CAD model of the bottle shown in the reference image. The model should capture the characteristic shape rather than simply approximating it with primitive solids.

The bottle should be rotationally symmetric. Use a sensible revolve or revolution-based workflow for the main body and create a hollow interior with an appropriate wall thickness.

DESIGN DIMENSIONS

FEATURE	DIMENSION
Overall height	250 mm
Maximum body diameter	65 mm
Neck outer diameter	28 mm
Bottle opening diameter	20 mm
Approx. wall thickness	2 mm
Base thickness	4 mm

MISSION 02

Submission

Submit the CAD model and enough information for us to verify that you understood and reproduced the reference.

PRACTICAL DETAILS

TIME	Approximately 1-2 hours
MODEL	STEP / STP file of the final CAD model
VIEW	One clean isometric screenshot of the completed model
SECTION	One section or side view showing the bottle profile and hollow interior
DRAWING	One simple dimensioned drawing or screenshot showing the main dimensions
SOFTWARE	Fusion 360, FreeCAD, SolidWorks, Onshape, or another CAD tool

WHAT WE ARE ACTUALLY LOOKING FOR

Can you observe a real object, break its geometry into simple features, and reproduce it as a clean CAD model? We are looking for sensible modelling, attention to proportions and dimensions, a clean feature order, and the ability to explain your choices.

You do not need advanced aerospace CAD knowledge. If something about the reference is unclear, make a reasonable assumption and state it.

CHECK BEFORE SUBMITTING

Overall height = 250 mm. Maximum body diameter = 65 mm. Neck outer diameter = 28 mm. Opening diameter = 20 mm. Approx. wall thickness = 2 mm. Base thickness = 4 mm.

The model is hollow, rotationally symmetric, visually consistent with the reference, and the feature tree is clean and logically organised.